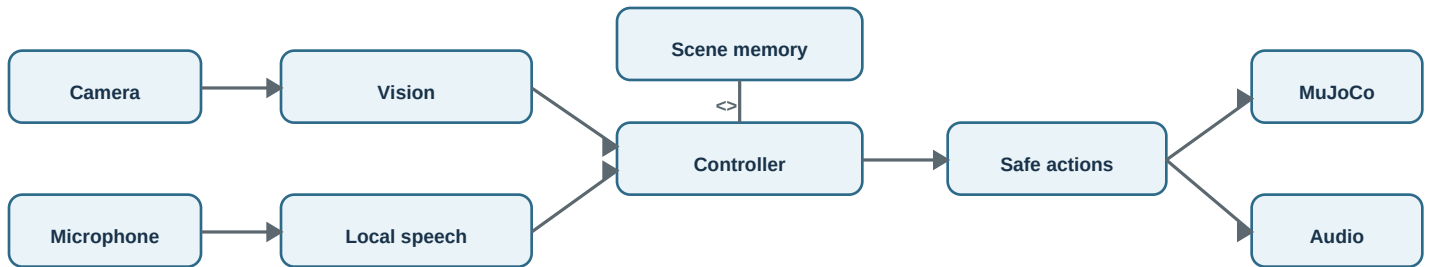


Live Character Robot

Technical note | End-to-end local character prototype

Architecture and data flow



Camera frames feed frontal-face engagement and HSV colored-object observation. Bounded microphone clips feed the CPU **faster-whisper base.en** model. Normalized text becomes a fixed motion command, scene-memory operation, or structured color goal. Only the deterministic controller writes joint state.

Protocol and model-to-action boundary

The internal protocol is a typed vocabulary: scene observations contain color, confidence, and horizontal position; goals contain a target color; actions are named primitives (**look-left**, **look-right**, **nod**, **shake**, **look-up**, **sleep**). Language never produces arbitrary joint values. The planner selects named actions, the motion layer converts named poses to ordered vectors, and every target is clamped to imported URDF limits before smoothstep interpolation.

For *inspect the blue object*, vision first searches for the requested color and estimates left/center/right position. The planner chooses a bounded orientation-and-nod sequence. The camera observes again after execution; the character reports completion only if the requested target remains visible.

Simulation, character, and deployment choices

MuJoCo provides quick URDF import, joint-limit inspection, and an interactive CPU-capable simulation. Speech recognition is local to avoid API cost and data transfer. macOS say or Ubuntu espeak-ng provides offline voice. Sound effects and music are synthesized at runtime; a warm shade-color pulse represents lamp light. The target is Ubuntu 24.04, Python 3.12, four CPU cores, and 8 GB RAM.

Data boundary: camera frames are processed in memory and discarded; recordings and raw measurement JSON remain local and Git-ignored; no cloud API is used.

Measured evidence

Development machine: Apple Silicon Mac, macOS 14.4.1

Metric	Result	Method
Speech-to-intent latency	0.7612 s mean	3 runs: 1.2071 / 0.5071 / 0.5695 s
Warm speech-to-intent	0.507-0.570 s	Runs after initial local model load
Peak resident memory	504.1 MiB	Process ru_maxrss
CPU use	143.1% of one core	3.2677 CPU s; about 1.4 cores average
Engagement accuracy	100.0%	360 frames across 3 guided trials

The transcript and resolved intent were correct in all runtime runs. Engagement trials used clear glasses and direct gaze followed by three disengagement variants: turning away, leaving frame, and looking down/elsewhere. Visible, absent, and overall frame accuracy were each 100%. This is a small controlled test, not a population-level reliability claim.

Reliability and physical reasoning

Engagement dwell times suppress detection flicker. Silence skips one turn; camera failure yields no invented observation; missing voice/audio preserves visual motion; microphone failure ends cleanly. The five-DOF action vocabulary is bounded and all targets respect imported joint limits. A physical robot would also require actuator feedback, velocity and acceleration enforcement, collision limits, an emergency stop, watchdogs, and a hardware-specific controller.

Known limitations

Perception: fixed HSV segmentation supports six colors and varies with exposure, lighting, shadows, reflections, and background. Pale or low-saturation objects and yellow/orange or blue/purple boundary hues may be misclassified. Frontal-face detection may degrade with glare, occlusion, or dim light. **Interaction:** fixed English phrases, five-second turns, one nonpersistent memory item, and no calibrated transform between camera and simulated lamp. **Simulation:** shade pulse is visual rather than photometric; no hardware dynamics or collision validation. **Deployment:** developed and measured on macOS; camera indexing, PipeWire audio, espeak-ng, OpenGL, and performance still require validation on the final Ubuntu laptop.

Ownership and scope

The prototype intentionally favors a smaller coherent local system over broad open-ended perception. AI-assisted development was used, while architecture, behavior, measurements, review, and final submission responsibility remain with the candidate.